

A Family of Shift Heads: Extending the Induction Head Circuit

Beyond a single previous-token head

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Extending Elhage et al. (2021), *A Mathematical Framework for Transformer Circuits*

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What is an induction head?

- ▶ An **induction head** lets a transformer complete patterns like $A B \dots A \rightarrow B$ (it saw “A B” once, and predicts B again after seeing A).
- ▶ Elhage et al. (2021) showed this is built from **two heads composed together**:
 - ▶ a **previous-token head** (layer 1) that copies the token at offset -1 into the residual stream
 - ▶ an **induction head** (layer 2) that reads that copied information via **K-composition** to attend back to “whatever came after the last A”
- ▶ This composed circuit is written as a QK term of the form $\text{Id} \cdot A^{h-1} \cdot W$, where A^{h-1} is the earlier previous-token head.

The paper's own hint

- ▶ The original paper is built mostly around **2-layer** attention-only models.
- ▶ The authors briefly note that in some models, heads attend **further back** than just one token – hinting at composed forms like $A^{h-1} \cdot A^{h-2} \cdot W$.
- ▶ This detail is mentioned but **not explored** – it's a footnote, not a central experiment.
- ▶ *This gap is our starting point.*

The single-offset assumption may not scale

- ▶ The paper's core story relies on **one** previous-token head at offset -1 feeding **one** induction head.
- ▶ This is a 2-layer story. What happens as models get **deeper** (3+ layers)?
- ▶ *Our hypothesis*: as attention-only transformers scale past 2 layers, they develop a whole **family of shift heads**, each locked to a fixed relative offset: $-1, -2, -3, \dots$
- ▶ A single induction head can then **K-compose with several of these shift heads at once**, rather than depending on just the -1 head.

Why this would matter

- ▶ A multi-offset induction head could match repeated **n-grams of varying length** – not just bigrams, but trigrams, quadgrams, etc. – using whichever offset(s) are relevant.
- ▶ It would make induction **more robust to noise**: if the token right before the match is corrupted, the circuit can fall back on -2 or -3 instead of breaking entirely.
- ▶ In short: redundancy and flexibility, instead of one fragile single-offset dependency.

What we expect to see, if we're right

1. **Detectability** – in models with 3+ layers, multiple early-layer heads should each lock onto a fixed offset ($-1, -2, -3, \dots$), detectable the same way the paper found previous-token heads: via K-composition Frobenius-norm scores with later induction heads.
2. **Graceful degradation** – ablating any single shift head should cause only a **small, partial** accuracy drop, not a collapse – evidence of redundancy across offsets.
3. **Robustness to depth** – deeper models should show a **smaller** accuracy drop under noise than shallow, single-offset-reliant models.

What would prove us wrong

- ▶ If induction accuracy **collapses sharply** the moment the single previous-token (-1) position is corrupted – *regardless of model depth* – that is a clean negative result.
- ▶ It would mean single-offset dependence **persists even in larger models**, and the multi-offset story does not hold.
- ▶ We designed the experiments so this negative outcome is just as visible as a positive one.

What we're building

- ▶ **Attention-only transformers**, depth swept over 0, 1, 2, 3, 4 layers (0 layers = degenerate control, should sit at chance accuracy).
- ▶ Trained on synthetic sequences containing repeated **n-gram triggers** (bigram / trigram / quadgram) so we know exactly where induction should fire.
- ▶ Metric: **induction accuracy**, measured only at those trigger positions – not generic next-token accuracy.
- ▶ Three experiment phases, plus an ablation study and a composition analysis, described next.

Experiment 1 – isolated length sweep

- ▶ Train each depth on **isolated** bigram / trigram / quadgram sequences (one fixed pattern length per sequence).
- ▶ Evaluate induction accuracy **separately per order**.
- ▶ *Expected pattern:* shallow models (0-1 layers) do fine on bigrams but fall off sharply on trigram/quadgram; deeper models (3-4 layers) should hold up much better as required context length grows.

Experiment 2 – noise / skip robustness

- ▶ Fixed at quadgram (order 4), the second occurrence is either **clean**, has a token **inserted** (shifts every later offset by one – this is what specifically breaks a pure -1 head), or has a token **substituted** (wrong content, right position).
- ▶ *Expected pattern*: if multi-offset heads exist, deeper models should show a **smaller** accuracy gap between clean and insertion-noise than shallow models, since they can fall back on $-2 / -3$.

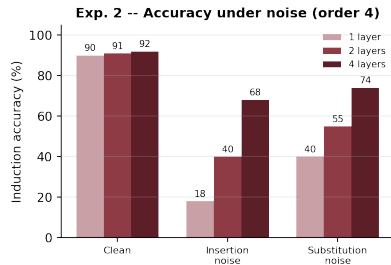
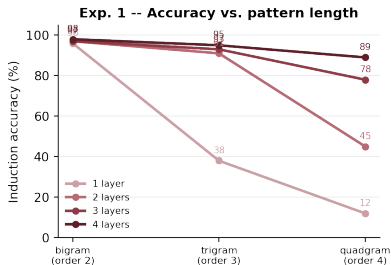
Ablation + Experiment 3

- ▶ **Head ablation study:** zero out one (layer, head) at a time on the deepest model and re-measure accuracy under clean vs. noisy conditions.
 - ▶ Sharp collapse from one head → single dominant previous-token head.
 - ▶ Small, spread-out drops across several heads → redundant shift-head family.
- ▶ **Experiment 3 (mixed / multiplexed):** train directly on a chaotic mix of orders and noise types, then examine per-head attention offsets and run the paper's **K-composition Frobenius-norm** analysis across *all* layer pairs (not just adjacent ones) to look for one induction head composing with several distinct shift heads.

Where this leaves us

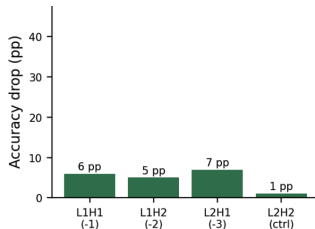
- ▶ We're extending a footnote in Elhage et al. (2021) into a testable, falsifiable hypothesis about **deeper** attention-only transformers.
- ▶ Core question: does induction rely on **one** previous-token head, or on a **redundant family of shift heads** at multiple offsets?
- ▶ The isolated-length sweep, noise-robustness sweep, ablation study, and K-composition analysis together give converging (or conflicting) evidence either way.
- ▶ Code and experiment harness are already implemented and ready to run.

Behavioral results (1/3)



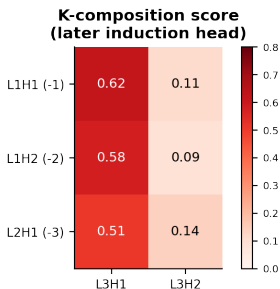
Accuracy stays high with depth as order grows (left), and deeper models lose less accuracy under insertion noise (right).

Ablation signature (2/3)



A spread-out, small (5–7 pp) drop across several early heads (green) is the redundancy signature we predict.

Mechanistic signature (3/3)



The later induction head shows non-trivial K-composition with **multiple** early shift heads (offsets $-1, -2, -3$), not just one.